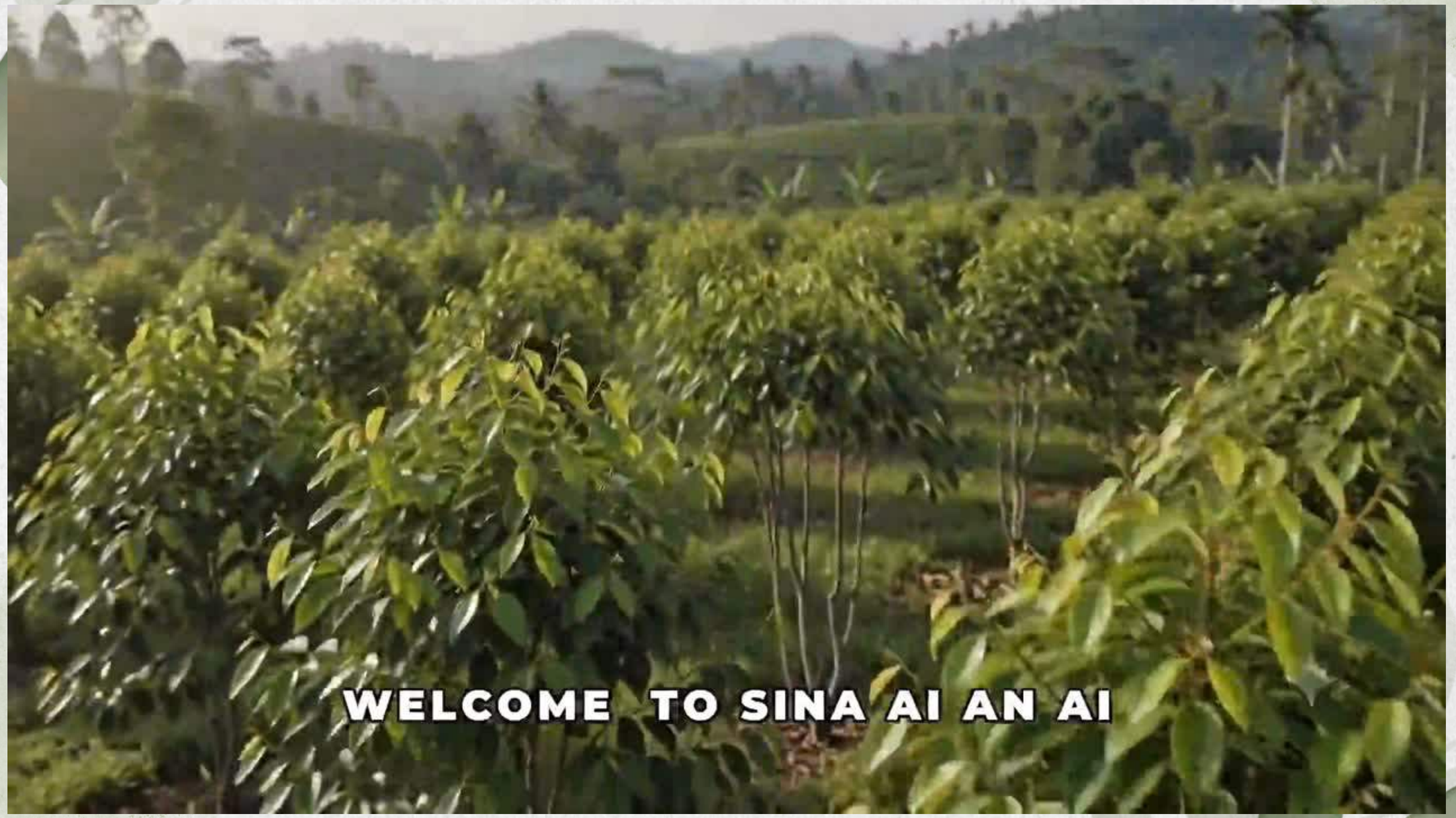
AI- DRIVEN SMART CINNAMON GROWTH MONITORING & AUTOMATED HARVESTING SYSTEM

R26-IT-102


WELCOME TO SINA AI AN AI

TEAM

Cinna AI



G.N RANAWEERA
IT22092656



WANASINGHE W.A.H.N
IT22092960



S.M.D.J.T.SAMARAWEERA
IT22065926

PROBLEM & MOTIVATION

Context

Cinnamon is one of Sri Lanka's most valuable export crops. However, several cultivation and harvesting activities still rely heavily on manual labour, experience and visual judgement.

Observed Problems

- Cinnamon harvesting requires highly skilled manual labour.
- Bark readiness and stem condition are often judged manually.
- Disease and insect damage can be difficult to identify at an early stage.
- Manual peeling can cause inconsistent bark quality and product wastage.
- Labour shortages affect harvesting efficiency and production capacity.
- Existing processes provide limited real-time monitoring and automated decision support.

SOLUTION OVERVIEW

Cinna AI is an integrated AI, IoT and robotic system designed to monitor cinnamon growth, identify plant health conditions and automate precision cinnamon bark harvesting.

Key Features

- Smart cinnamon growth and environmental monitoring
- AI-based disease and insect damage detection
- Harvest readiness prediction
- Cinnamon bark boundary detection using computer vision
- Automatic harvesting coordinate generation
- Intelligent robotic peeling control
- Bark length and width measurement
- Real-time machine monitoring and control
- Safety controls including Start, Stop and Emergency Stop

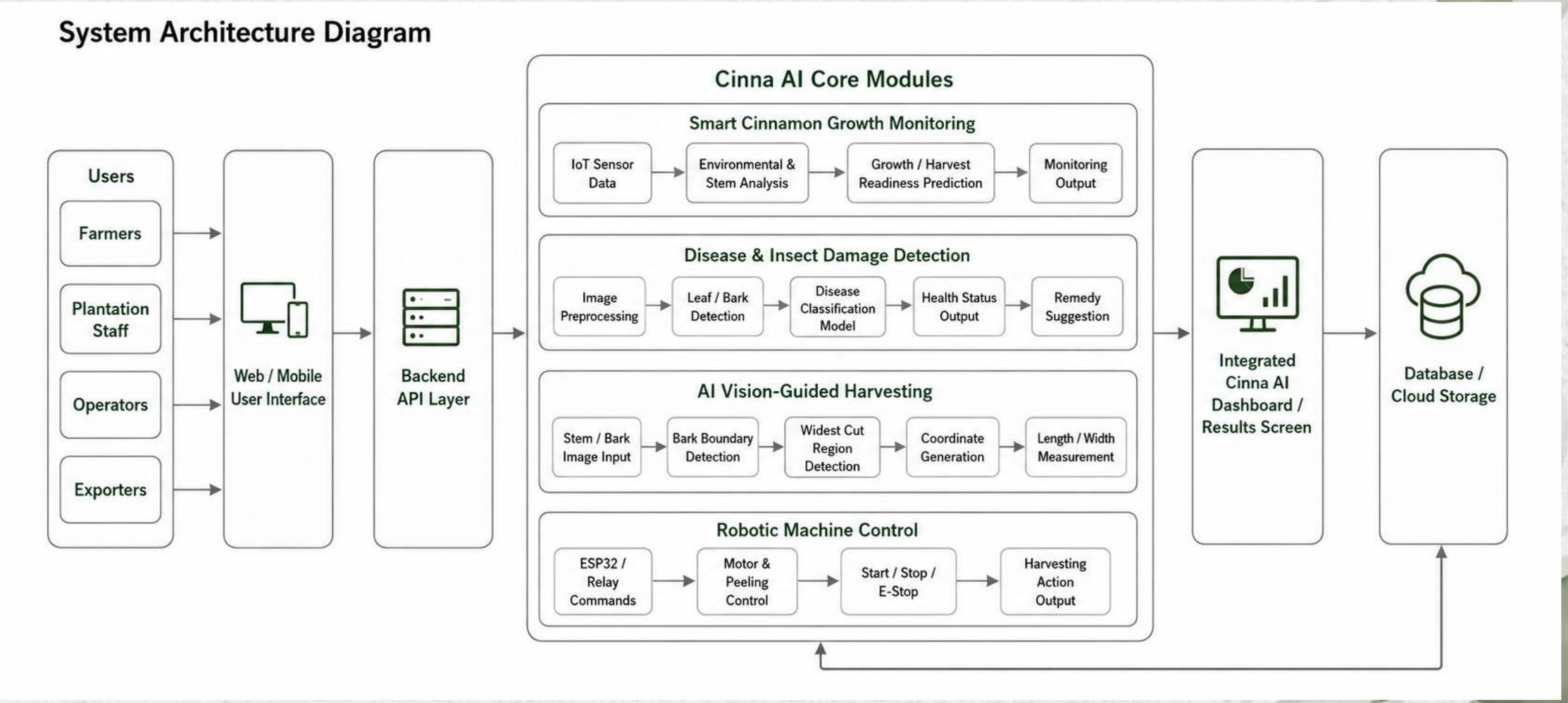
Non-Functional Goals

- High detection accuracy
- Fast AI processing
- Easy-to-use web interface
- Safe robotic operation
- Reliable real-time communication

OVERALL RESEARCH GAP

Feature	Plantix App	Agrio App	Referring to Research [1]	Referring to Research [2]	Suggested Solution (Cinna AI)
Cinnamon-specific intelligent monitoring system	✗	✗	✗	✗	✓
Disease & insect damage detection through image capture	✓	✓	✗	✗	✓
Growth and harvest readiness prediction	✗	✗	✓	✗	✓
IoT-based environmental and stem monitoring	✗	✗	✗	✓	✓
Cinnamon bark boundary detection using AI vision	✗	✗	✗	✗	✓
Automatic harvesting coordinate generation	✗	✗	✗	✗	✓
AI-guided robotic bark harvesting / peeling	✗	✗	✗	✓	✓
One integrated AI + IoT + Robotics platform	✗	✗	✗	✗	✓

SYSTEM ARCHITECTURE & INTEGRATION



SYSTEM ARCHITECTURE & INTEGRATION

Components

Component 01- Smart Cinnamon Growth Monitoring & Harvest Readiness Prediction- IT22092960
Component 02- Cinnamon Disease and Insect Damage Detection- IT22092656
Component 03- AI Vision-Guided Robotic Cinnamon Bark Harvesting- IT22065926

Integration Points

- REST API communication
- JSON-based data exchange
- IoT sensor data transmission
- Image uploads for AI analysis
- AI-generated harvesting coordinates
- ESP32 communication
- Relay-based motor control
- Shared database
- Integrated Cinna AI dashboard

COMMERCIALIZATION

Target Users / Market – Cinnamon farmers, plantations, processing centers and exporters in Sri Lanka.

Value Proposition – AI-driven smart cinnamon monitoring and robotic harvesting system that improves accuracy, reduces manual labour and supports better harvesting decisions.

Cost & Feasibility – Built using open-source AI, affordable IoT sensors, ESP32 controllers and cloud technologies, making it practical for commercial deployment.

Adoption Plan – Prototype testing → Plantation field trials → Performance improvement → Industry partnerships → Commercial deployment.

THANK YOU

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